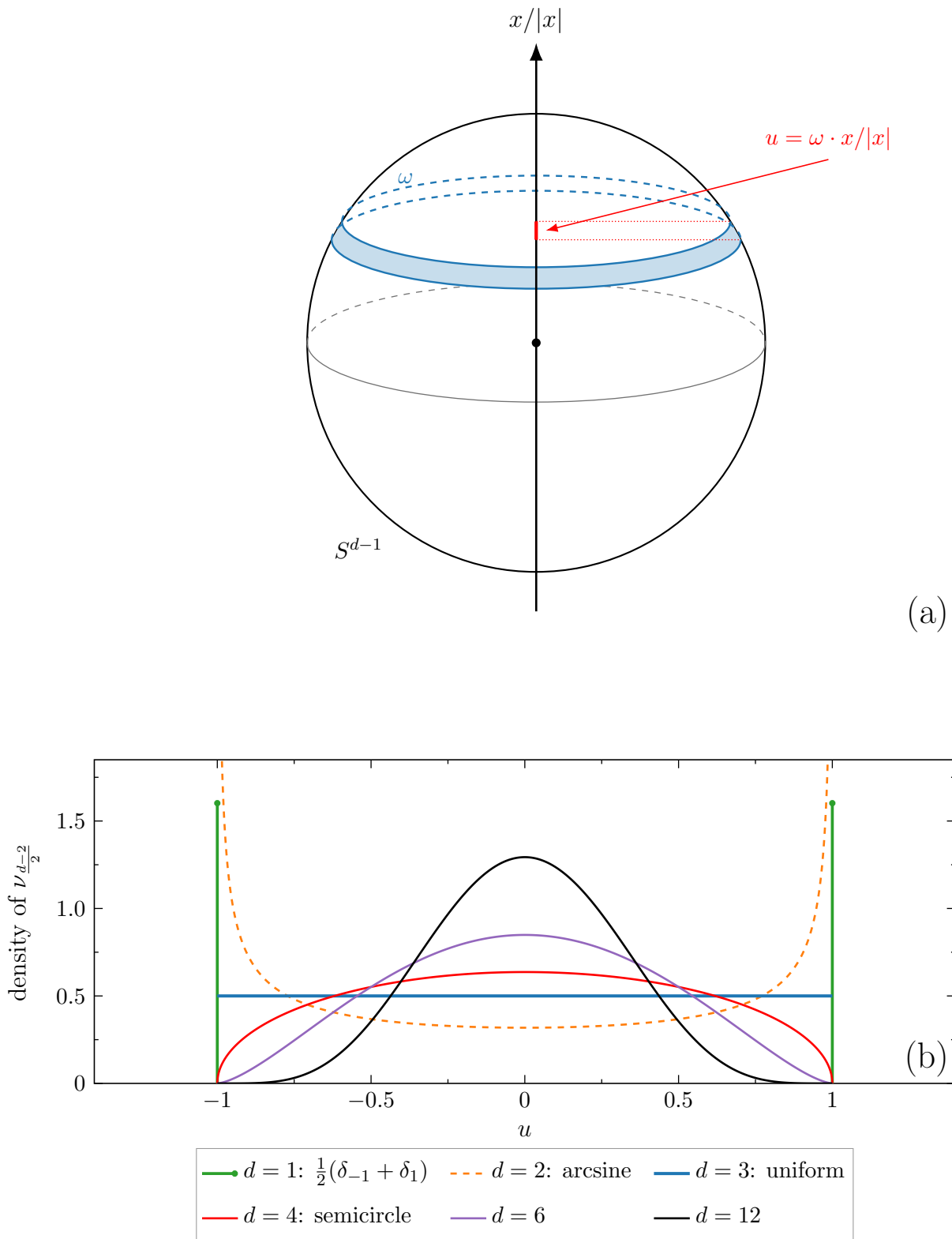

$x/|x|$
$u = \omega \cdot x/|x|$
ω
S^{d-1}
(a)
density of $\nu_{\frac{d-2}{2}}$
1.5
1.0
0.5
0
−1
−0.5
0
0.5
1
u
(b)
$d = 1$: $\frac{1}{2}(\delta_{-1} + \delta_1)$
$d = 2$: arcsine
$d = 3$: uniform
$d = 4$: semicircle
$d = 6$
$d = 12$